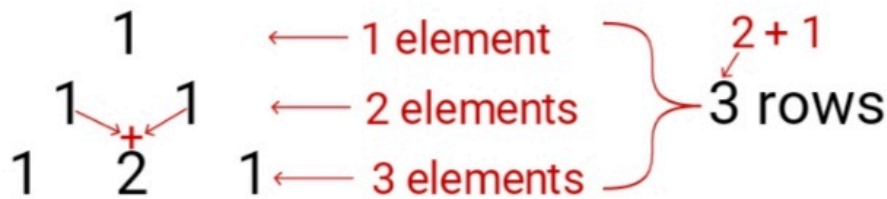


- The number of rows equals $n + 1$ (where n is the power to which the bracket is raised).
- The first row has got one term and the number of terms increases from 1 row to the next by 1.
- The first and last element in a row is always 1.
- The other elements are filled by adding the top 2 terms e.g for $n = 2$:



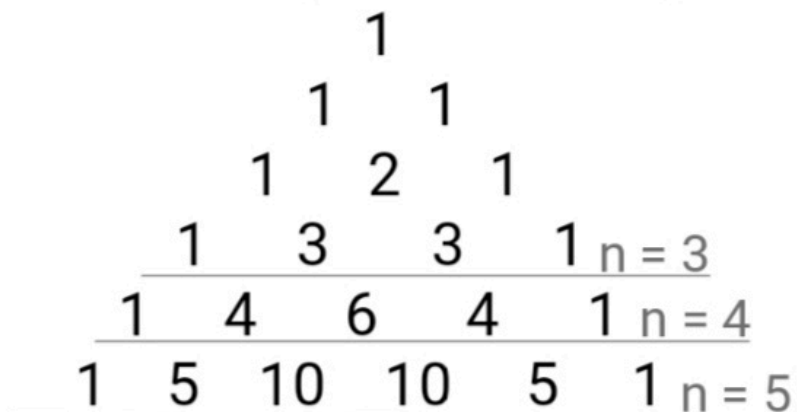
- Here's a clean one :



- For $n = 4$:



- You can just draw the triangle for a higher power e.g for $n = 5$ and the same can still be used for lesser powers by just referring to the corresponding row :



$$(a + b)^n = z_1 \cdot a^{y_1} \cdot b^{m_1} + z_2 \cdot a^{y_2} \cdot b^{m_2} + \dots + z_{n+1} \cdot a^{y_{n+1}} \cdot b^{m_{n+1}}$$

- There will be $n + 1$ elements adding up each other.
- $z_1, z_2, \dots, z_{n+1}$ are the numbers from the corresponding row in the triangle (from the first(1) until the last(1 again))
- z_1 and z_{n+1} are therefore always 1.
- $y_1, y_2, \dots, y_{n+1}$ are $n, n - 1, \dots, 0$.
- $m_1, m_2, \dots, m_{n+1}$ are $0, 0 + 1, \dots, n$.

Examples

1. Expand the following brackets

(a) $(2x + 3)^3$

(b) $(3x - 2)^4$

(c) $(\frac{1}{2}x + 4)^5$

sol:

1. Let's just draw the triangle for $n = 5$ (the highest power) and we will use it in all cases just by referring to the correct row :

$$\begin{array}{ccccccc}
 & & 1 & & & & \\
 & 1 & & 1 & & & \\
 & 1 & 2 & 1 & & & \\
 1 & 3 & 3 & 1 & n = 3 & & \\
 1 & 4 & 6 & 4 & 1 & n = 4 & \\
 1 & 5 & 10 & 10 & 5 & 1 & n = 5
 \end{array}$$

$$(a + b)^n = z_1 \cdot a^n \cdot b^0 + z_2 \cdot a^{n-1} \cdot b^1 + \dots + z_{n+1} \cdot a^0 \cdot b^n$$

OR

$$(a + b)^n = a^n + z a^{n-1} b + \dots + b^n \text{ since } z_1 = 1, z_{n+1} = 1, b^0 = 1 \text{ and } a^0 = 1$$

$$\begin{aligned}
 (a) (2x + 3)^3 &= (1)(2x)^3(3)^0 + 3(2x)^2(3)^1 + 3(2x)(3)^2 + (1)(2x)^0(3)^3 \\
 &= 8x^3 + 3(4x^2)(3) + 3(2x)(9) + 9 \\
 &= 8x^3 + 36x^2 + 54x + 9
 \end{aligned}$$

$$\begin{aligned}
 (b) (3x + 2)^4 &= (3x)^4 + 4(3x)^3(-2) + 6(3x)^2(-2)^2 + 4(3x)(-2)^3 + (-2)^4 \\
 &= 81x^4 + 4(27x^3)(-2) + 6(9x^2)(4) + 4(3x)(-8) + 16 \\
 &= 81x^4 - 216x^3 + 216x^2 - 96x + 16
 \end{aligned}$$

$$\begin{aligned}
 (c) \left(\frac{1}{2}x + 4\right)^5 &: \\
 &= \left(\frac{1}{2}x\right)^5 + 5\left(\frac{1}{2}x\right)^4(4) + 10\left(\frac{1}{2}x\right)^3(4)^2 + 10\left(\frac{1}{2}x\right)^2(4)^3 + 5\left(\frac{1}{2}x\right)(4)^4 + 4^5 \\
 &= \frac{1}{32}x^5 + 5\left(\frac{1}{16}x^4\right)(4) + 10\left(\frac{1}{8}x^3\right)(16) + 10\left(\frac{1}{4}x^2\right)(64) + 5\left(\frac{1}{2}x\right)(256) + 1024 \\
 &= \frac{1}{32}x^5 + \frac{5}{4}x^4 + 20x^3 + 160x^2 + 640x + 1024
 \end{aligned}$$

Practice

1) Expand the following brackets

(a) $\left(\frac{1}{3}x + 4\right)^4$

(b) $(x^2 - 1)^3$

(c) $(2x + 1)^5$

sol:

$$1(a) \left(\frac{1}{3}x + 4\right)^4 = \frac{1}{81}x^4 + \frac{16}{27}x^3 + \frac{32}{3}x^2 + \frac{256}{3}x + 256$$

$$(b) (x^2 - 1)^3 = x^6 - 3x^4 + 3x^2 - 1$$

$$(c) (2x + 1)^5 = 32x^5 + 144x^4 + 80x^3 + 40x^2 + 10x + 1$$

The Binomial theorem

- The Pascal's triangle can only be used for the expansion of brackets which are raised to a positive integral power.
- The Binomial theorem can however be used for the expansion of brackets that are raised to positive integral powers, negative integral powers or even fractional powers.

1. A bracket raises to a positive integral power.

- Let's consider the bracket $(2 + x)^3$

By Pascal

$$\begin{array}{cccc} & & 1 & \\ & 1 & & 1 \\ & 1 & 2 & 1 \\ 1 & 3 & 3 & 1 \end{array}$$

$$\begin{aligned}(2 + x)^3 &= (1)(2^3)(x)^0 + 3(2^2)(x)^1 + 3(2^1)(x^2) + (1)(2^0)(x^3) \\ &= 2^3 + 3(4)x + 3(2)x^2 + x^3 \\ &= 8 + 12x + 6x^2 + x^3\end{aligned}$$

Binomial _ _ _

Note : - Binomial only expands brackets of the form $(a + x)^n$, where **a** equals 1.

- So on the bracket $(2 + x)^3$, we will first divide each term in the bracket by 2 so that we will be left with a bracket of the form $(1 + x)^3$

$$\begin{aligned}(2 + x)^3 &= [2(1 + \frac{x}{2})]^3 \\ &= 2^3(1 + \frac{x}{2})^3\end{aligned}$$

$$= 8\left(1 + \frac{x}{2}\right)^3$$

This is the bracket which we are supposed to expand.

$$(a + x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$$

$$\binom{n}{x} = \frac{n \times (n-1) \times (n-2) \times \dots}{x!}$$

x times

Examples

$$\binom{5}{2} = \frac{5 \times 4}{2!} = 20$$

On calculator is punched
as : - 5 C 2
- 4 C 3

$$\binom{4}{3} = \frac{4 \times 3 \times 2}{3!} = 4$$

Factorials :

- 2! is read as 2 factorial.

$$1! = 1$$

$$2! = 2 \times 1 = 2$$

$$3! = 3 \times 2 \times 1 = 6$$

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

Back to our story :

$$(a + x)^n = 1 + \binom{n}{1}x + \binom{n}{2}x^2 + \binom{n}{3}x^3 + \dots + \binom{n}{n}x^n$$

where $a = 1$

$$\begin{aligned} \therefore (2 + x)^3 &= 8\left(1 + \frac{x}{2}\right)^3 \\ &= 8\left[1 + \binom{3}{1}\left(\frac{x}{2}\right) + \binom{3}{2}\left(\frac{x}{2}\right)^2 + \binom{3}{3}\left(\frac{x}{2}\right)^3\right] \\ &= 8\left[1 + \frac{3}{2}x + \frac{3}{4}x^2 + \frac{1}{8}x^3\right] \end{aligned}$$

$$= 8 + 12x + 6x^2 + x^3 \} \text{ same as the result obtained using Pascal}$$

2. Bracket raised to a fractional power or a negative integer

- Pascal haichaite apa and so tinoshandisa Binomial chete.

$$(a + x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)x^3}{3!} + \dots + x^n$$

Note: - In this case expansion yacho haigume so tinongoiratidza hedu kusvikira paterm ina x^3 then tototi ' + _____ + x^n '

Examples

1. Expand the following brackets upto and a term including x^3

(a) $(1 + x)^{\frac{1}{2}}$

(b) $(1 - 2x)^{-3}$

sol:

1(a) a yagara itori 1 saka let's worry less.

$$\begin{aligned} (1 + x)^{\frac{1}{2}} &= 1 + \frac{\frac{1}{2}x}{1!} + \frac{\frac{1}{2}(\frac{1}{2} - 1)x^2}{2!} + \frac{\frac{1}{2}(\frac{1}{2} - 1)(\frac{1}{2} - 2)x^3}{3!} + \dots + x^{\frac{1}{2}} \\ &= 1 + \frac{\frac{1}{2}x}{1} + \frac{\frac{1}{2}(-\frac{1}{2})x^2}{2} + \frac{(-\frac{1}{2})(-\frac{3}{2})x^3}{6} + \dots + x^{\frac{1}{2}} \\ &= 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3 + \dots + x^{\frac{1}{2}} \end{aligned}$$

$$\begin{aligned}
 (b) \quad (1 - 2x)^{-3} &= 1 + \frac{(-3)(-2x)}{1!} + \frac{(-3)(-3-1)(-2x)^2}{2!} + \frac{(-3)(-3-1)(-3-2)(-2x)^3}{3!} + \dots \\
 &\quad \dots + (-2x)^{-3} \\
 &= 1 + 6x + 12(4x^2) - 60(-8x^3) + \dots + \frac{1}{(-2x)^3} \\
 &= 1 + 6x + 48x^2 + 480x^3 + \dots - \frac{1}{8x^3}
 \end{aligned}$$

Practice

1) Expand the following brackets upto and a term including x^3

(a) $(2 - x)^{-2}$

(b) $(3 + x)^{-1}$

Ans:

$$1(a) \quad (2 - x)^{-2} = \frac{1}{4} + \frac{1}{4}x + \frac{3}{16}x^2 + \frac{1}{8}x^3 + \dots + \frac{1}{x^2}$$

$$(b) \quad (3 + x)^{-1} = \frac{1}{3} - \frac{1}{9}x + \frac{1}{27}x^2 - \frac{1}{81}x^3 + \dots + \frac{3}{x}$$

SOME PAST EXAM QUESTIONS

ZIMSEC 6042/1 J 2020

5(a) Expand $(p - x)^{-2}$ as a series of ascending powers of x upto a term in x^3 where p is a positive constant. [5]

(b) Given that the coefficient of x^2 in the expansion is $\frac{3}{16}$, find the value of p . [2]

(c) Hence state the set of values of x for which the expansion is valid. [1]

sol :

$$\begin{aligned}5(a) \quad (p - x)^{-2} &= [p(1 - \frac{x}{p})]^{-2} \\&= p^{-2}(1 - \frac{x}{p})^{-2} \\&= \frac{1}{p^2}(1 - \frac{x}{p})^{-2}\end{aligned}$$

$$(1 + x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)x^3}{3!}$$

$$\begin{aligned}\therefore (1 - \frac{x}{p})^{-2} &= 1 + \frac{(-2)(-\frac{x}{p})}{1!} + \frac{(-2)(-2-1)(-\frac{x}{p})^2}{2!} + \frac{(-2)(-2-1)(-2-2)(-\frac{x}{p})^3}{3!} \\&= 1 + \frac{\frac{2}{p}x}{1} + \frac{6(\frac{x^2}{p^2})}{2} + \frac{(-24)(\frac{-x^3}{p^3})}{6} \\&= 1 + \frac{2}{p}x + \frac{3}{p^2}x^2 + \frac{4}{p^3}x^3\end{aligned}$$

$$\therefore (p - x)^{-2} = \frac{1}{p^2} [1 + \frac{2}{p}x + \frac{3}{p^2}x^2 + \frac{4}{p^3}x^3]$$

$$= \frac{1}{p^2} + \frac{2}{p^2}x + \frac{3}{p^4}x^2 + \frac{4}{p^5}x^3$$

$$(b) \quad \frac{3}{p^4} = \frac{3}{16}$$

$$\begin{aligned}16(\cancel{3}) &= \cancel{3}(p^4) \\2^4 &= p^4\end{aligned}$$

$$\therefore p = 2$$

(c) - For a bracket of $(a + x)^n$, a must be 1 and $|x| < 1$
- Here we've got $\frac{1}{p^2}(1 - \frac{x}{p})^{-2}$

$$\therefore \left| \frac{x}{p} \right| < 1 \quad \text{But } p = 2$$

$$\therefore \left| \frac{x}{2} \right| < 1$$

$$\left(\frac{x}{2} \right)^2 < 1^2$$

$$\frac{x^2}{4} < 1$$

$$x^2 < 4$$

$$\text{For } x^2 = 4, x = \pm 2$$

Note: - For $x^2 < a$, $-\sqrt{a} < x < \sqrt{a}$

- For $x^2 > a$, $x < -\sqrt{a} \cup x > \sqrt{a}$

$$\therefore \text{For } x^2 < 4, -2 < x < 2$$

A'level Sciences tutor : *Sir Cosy Chikamhi*

- Pure Mathematics
- Physics
- Chemistry

ZIMSEC/CAMBRIDGE SYLLABUS

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Revelation 16:15 (KJV)

Behold, I come as a thief. Blessed is he that watcheth, and keepeth his garments, lest he walk naked, and they see his shame.